

Project Apache Hadoop Yarn

Task: Component “Resource manager scheduler”

Scenario: As a developer delving into Apache Hadoop Yarn internals, your task is to understand the design decisions behind the Resource manager (capacity) scheduler component, including capacities. You aim to learn its functionality and grasp its inner workings from components, connectors, and architectural solutions.

Existence

Question 1: What are the components and connectors of the resource manager (capacity) scheduler? Briefly name and describe each component, determine their type (physical or logical), determine how they are connected, and how they achieve the requirements of resource manager (capacity) scheduler.

Property

Question 2: What quality attributes and quality attribute solutions (e.g., patterns or tactics) are considered in the design of resource manager (capacity) scheduler? Mention the considered quality attributes and describe the solutions applied, and how they achieve the requirements.

Executive

Question 4: What are the technologies applied in the resource manager (capacity) scheduler, and what is the rationale behind selecting these technologies? Mention the technologies and tools, their associated components, rationale, and how they achieve the requirements.

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Task: Requirement Performance

Scenario: Performance is a key feature of Hadoop Yarn. Understanding how this feature is designed in Yarn will provide insight into how data stored with Yarn can be efficiently retrieved. You are tasked with exploring the architectural decisions that enable high performance of Hadoop Yarn capacity scheduler.

Existence

Question 1: What components and connectors in Yarn resource manager (capacity) scheduler contribute to its performance? Briefly name and describe each component, determine their type (physical or logical), determine how they are connected, and how they achieve the performance requirements of resource manager (capacity) scheduler.

Property

Question 2: What solutions (e.g., patterns or tactics) are implemented to support the performance quality attribute in the Yarn resource manager (capacity) scheduler? Mention and describe the solutions applied, and how they achieve the requirements.

Executive

Question 3: Which tools and technologies are integrated into Yarn resource manager (capacity) scheduler to ensure performance? Mention the technologies and tools, their associated components, rationale, and how they achieve the requirements.